

SMART INDIA HACKATHON 2025



Problem Statement ID : SIH25042

Problem Statement Title : Identifying Taxonomy and Assessing Biodiversity from eDNA Datasets

Theme : Miscellaneous

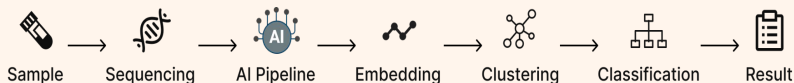
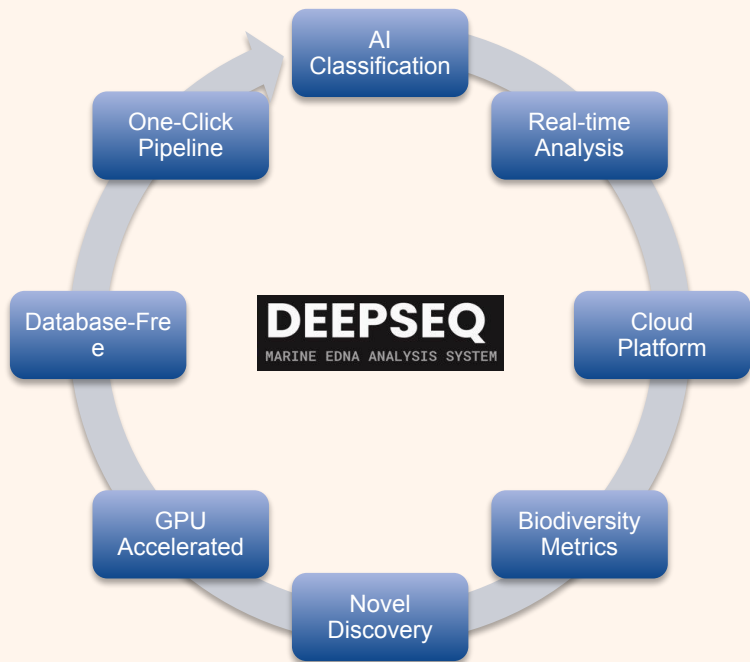
PS Category : Software

Team ID : 77056

Team Name : STORM SURGE



Storm Surge



IDEA Title

Currently Faced Problems :

- ✓ **Incomplete databases**, causing misclassification and unidentified species
- ✓ **Slow, manual pipelines**, taking weeks to process eDNA data
- ✓ **High computational costs**, requiring expensive resources
- ✓ **Fragmented workflows**, with no unified analysis platform
- ✓ **Limited accessibility**, excluding smaller research teams

Our Idea :

- ✓ **AI-based classification**, independent of incomplete databases
- ✓ **Faster processing**, reducing analysis time from weeks to hours
- ✓ **Novel species discovery**, through unsupervised learning
- ✓ **Real-time biodiversity insights**, with abundance and diversity metrics
- ✓ **Unified platform**, for upload, processing, and visualization
- ✓ **Accessible to all**, enabling non-experts to analyze eDNA data



[Hosted Website](#)
(click here)

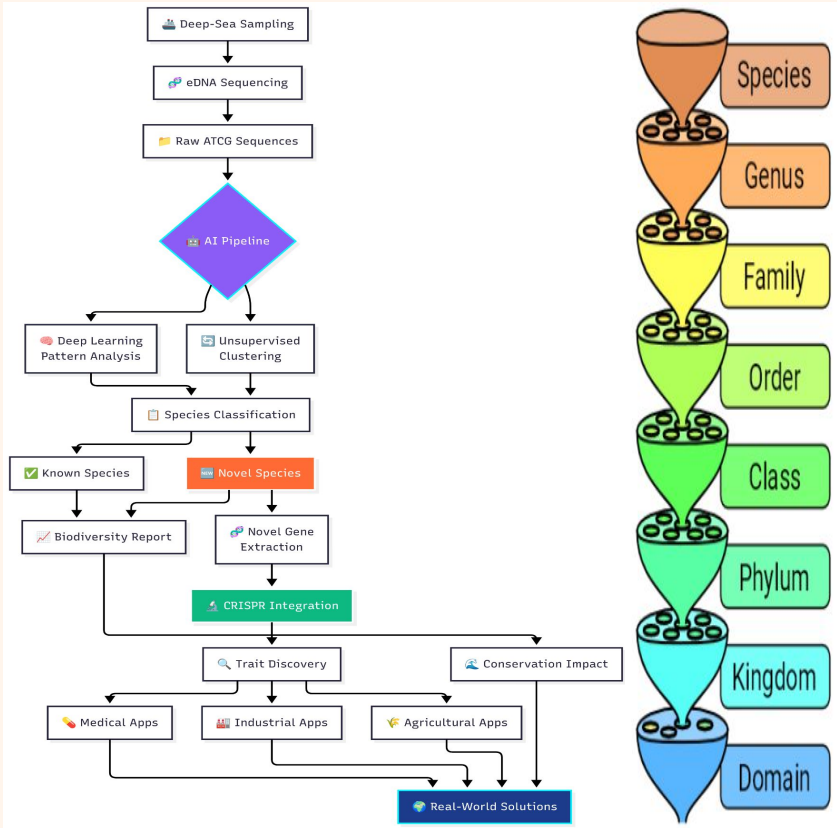
[Youtube](#)
(click here)

[Source Code](#)
(click here)

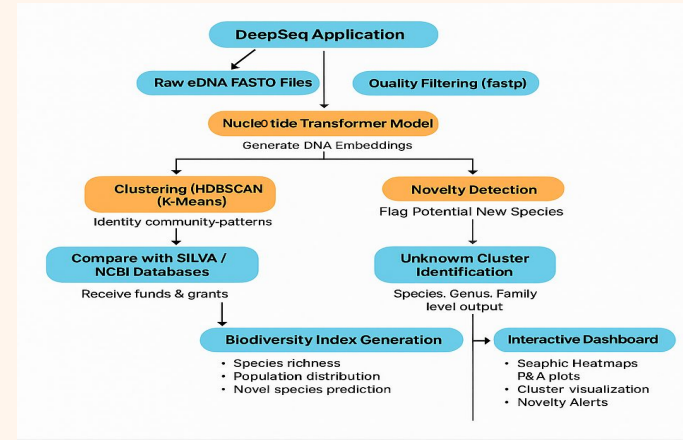
[Figma](#)
(click here)

Storm Surge

TECHNICAL APPROACH



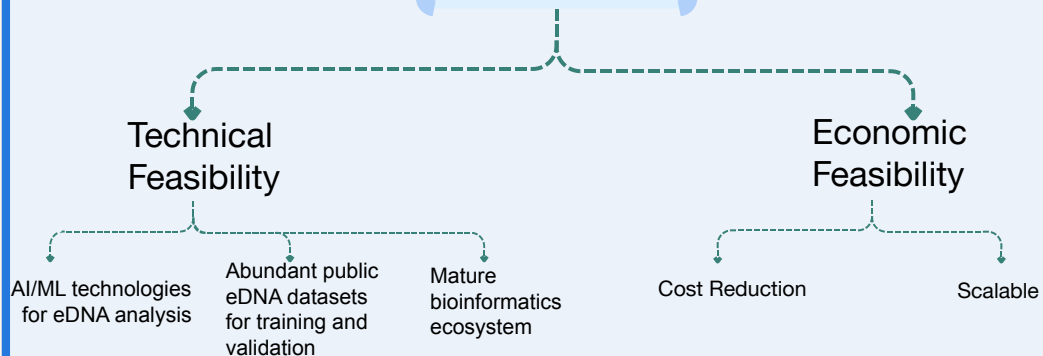
Flow of project



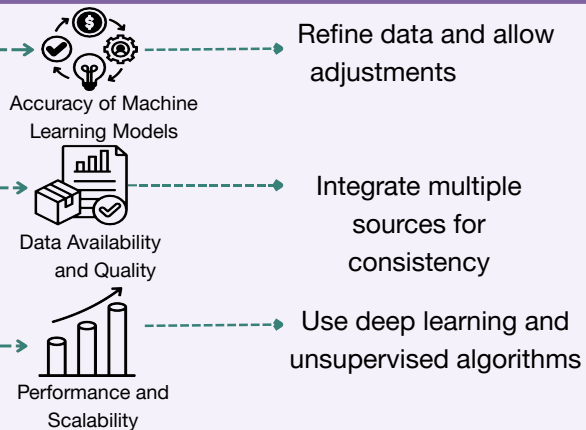
Foundation of eDNA-Based Taxonomy



Feasibility



Challenges



Our Strategic Approach

Learn DNA Patterns instead
Patterns Instead of Matching
Databases

Transformer-based sequence
embeddings

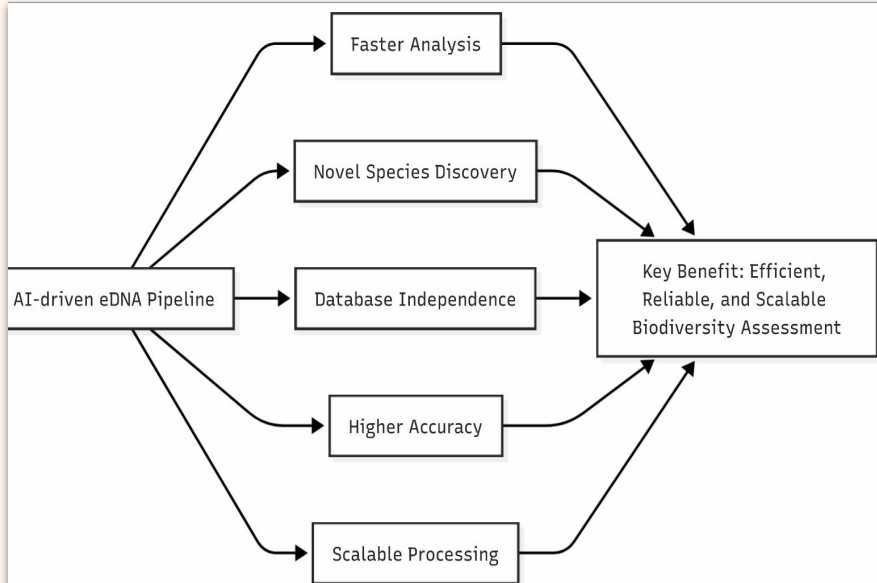
Pretrain
+ fine-tune on 18S/ COI

Unsupervised discovery of
novel taxa

Alignment- free classification

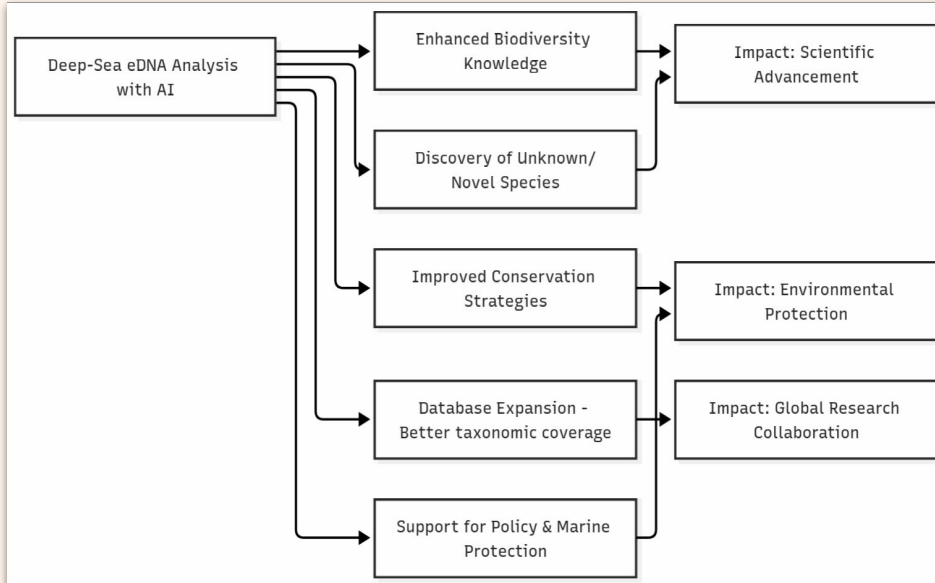
AI-based abundance
estimation

Solution Benefits

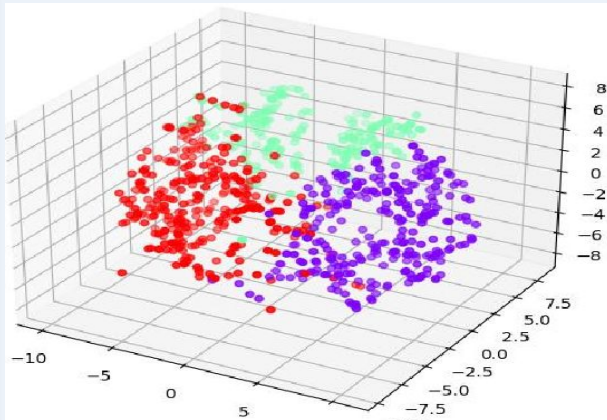


- Ensures **faster, scalable, and database-independent** biodiversity analysis using AI.
- Enables **novel species discovery** and delivers **higher accuracy** in taxonomic classification.

Target Audience Impacts



- Supports **scientific advancement, environmental protection, and policy making**.
- Drives **global collaboration** through better biodiversity data and novel discoveries.



Objectives

- Accurate Taxonomic Identification
- Detection of Novel Species
- Faster AI-Based Processing
- High-Quality eDNA Data Interpretation
- Scalable System for Large Datasets



Data Preparation

- Obtain Raw eDNA FASTQ Files
- Perform Quality Filtering (fastp)
- Remove Low-Quality Reads
- Extract Target Marker Genes (18S rRNA / COI)
- Prepare Clean Input Data for Model



Model Processing

- Generate DNA Embeddings using Nucleotide Transformer (DNABERT-2)
- Apply Clustering Algorithms (HDBSCAN)
- Group Sequences by Biological Similarity
- Run Novelty Detection for Unknown Species
- Compare Outputs with SILVA / NCBI Database



Analysis & Interpretation

- Assign Taxonomy (Species → Genus → Family)
- Identify Community Structure
- Quantify Biodiversity Metrics
- Detect Rare / Novel Species
- Generate Graphs & PCA Visualizations



Dashboard & Insights

- Build Interactive UI for Scientists
- Display Heatmaps, Clusters, and Species Lists
- Show Biodiversity Index Metrics
- Provide Novel Species Alerts
- Allow CSV/JSON Export for Researchers



Deployment

- Deploy AI Model on Cloud (AWS / GCP)
- Set Up API for File Upload & Processing
- Real-Time Monitoring & Logs
- Secure Authentication for Research Teams
- Scale System for National-Level Usage

■ Nucleotide transformer based Approaches with contrastive Learning

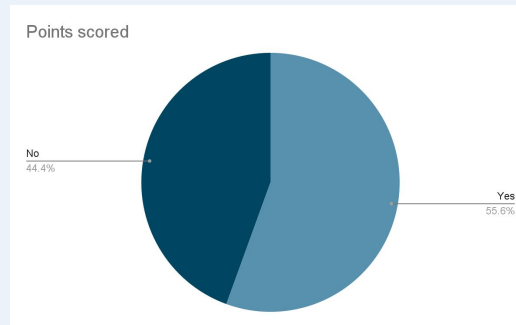
- Reference: <https://www.nature.com/articles/s41592-024-02523-z>
- Reference: https://hdbscan.readthedocs.io/en/latest/how_hdbscan_works.html
- Reference: <https://www.arxiv.org/pdf/2509.25274>

■ Biodiversity Informatics & Global Databases

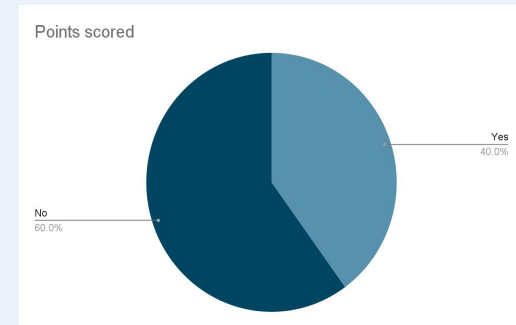
- Reference: <https://www.ncbi.nlm.nih.gov/>
- Reference: <https://www.arb-silva.de/>

Survey Questions

Do you think monitoring biodiversity is important for protecting ecosystems?



Do you think eDNA can help detect endangered or invasive species early?

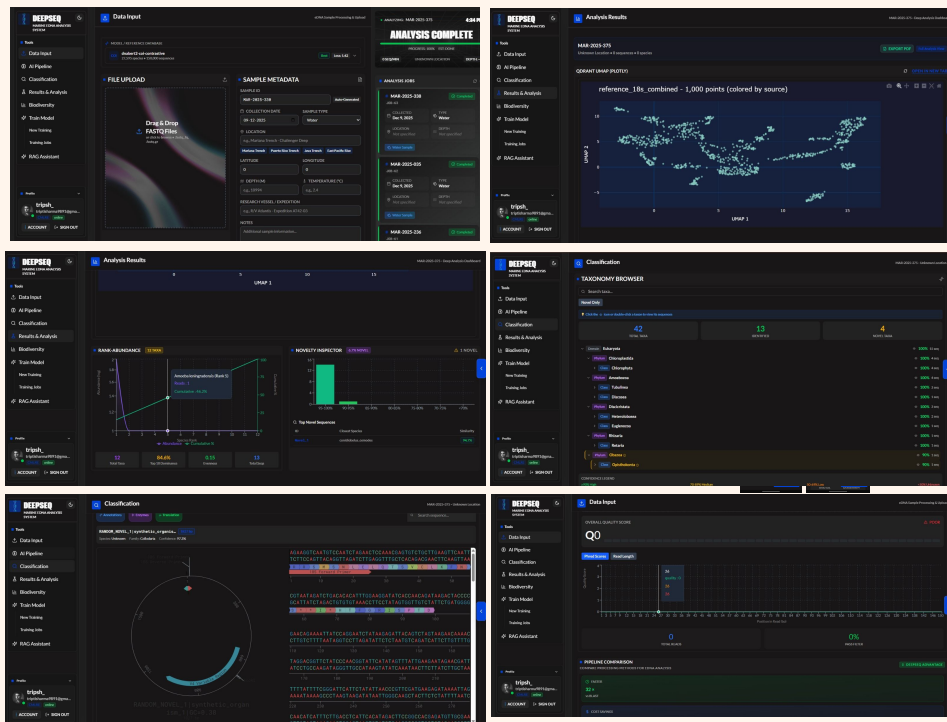


Storm Surge

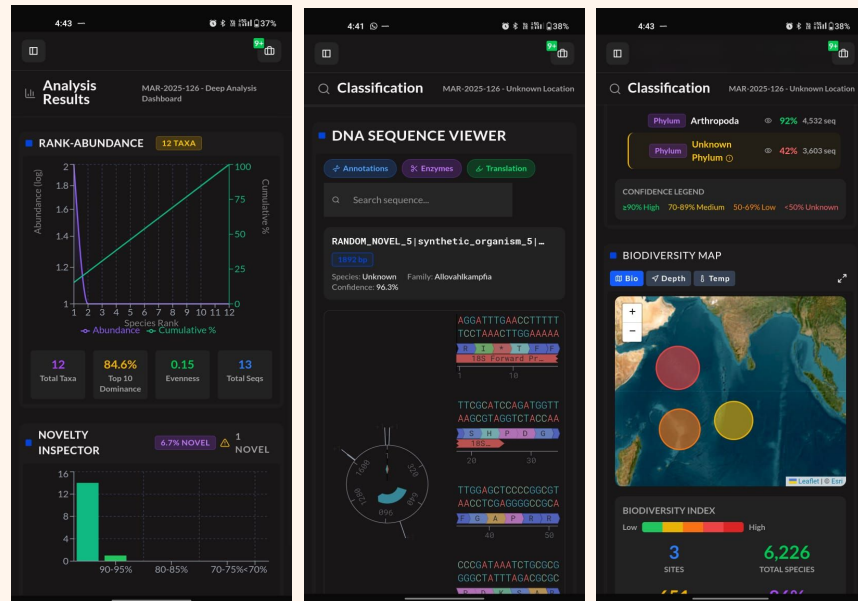
UI SCREENS

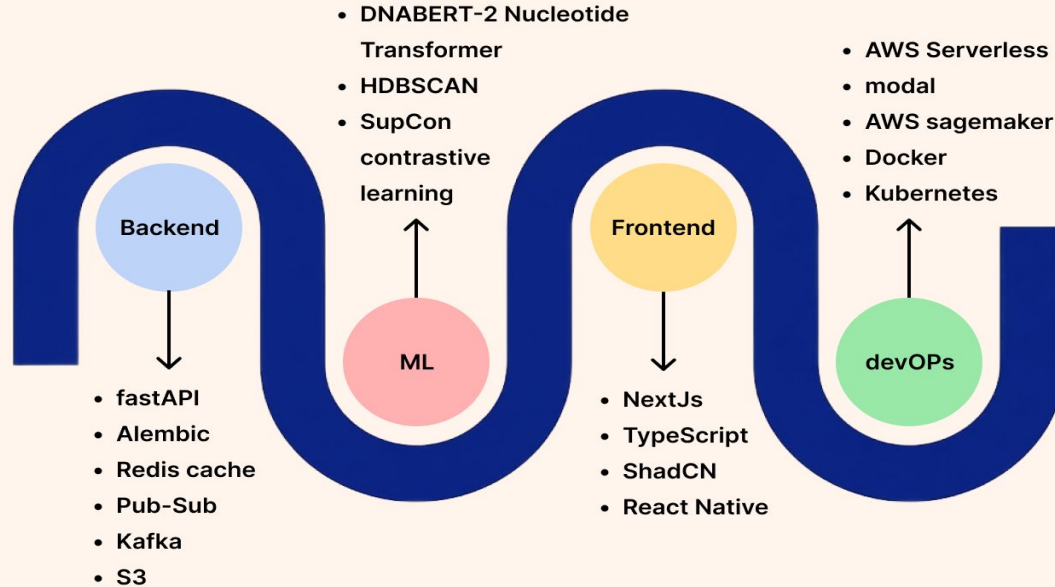


Website



APP





Team Leader

ANISH SINGH CHAUHAN

B.Tech- CSE(DS)

3rd Year

Domain: Backend Developer

Team Member

TRIPTI SHARMA

B.Tech- CSIT

3rd Year

Domain: UI/UX Designer

Team Member

KARTIKAY SINGH

B.Tech- CSE(AIML)

3rd Year

Domain: Frontend Developer

Team Member

APURVI KANAUIA

B.Tech- IT

3rd Year

Domain: UX Designer

Team Member

NAVYA GUPTA

B.Tech- CSE

3rd Year

Domain: Frontend Developer

Team Member

DIVYANSH VERMA

B.Tech- CSE

3rd Year

Domain: Backend Developer